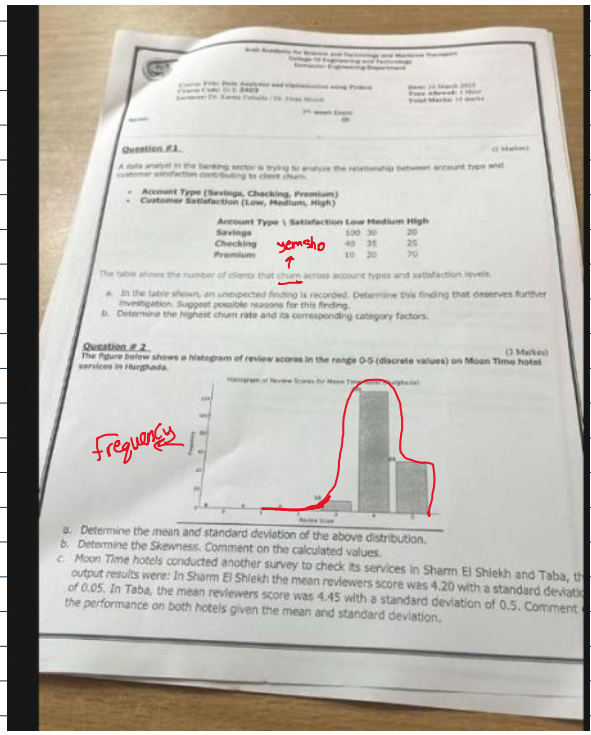


Question #1

(2 Marks)

A data analyst in the banking sector is trying to analyze the relationship between account type and customer satisfaction contributing to client churn.

- **Account Type (Savings, Checking, Premium)**
- **Customer Satisfaction (Low, Medium, High)**

Account Type \ Satisfaction	Low	Medium	High
Savings	100	30	20
Checking	40	35	25
Premium	10	20	70

yemsho

The table shows the number of clients that churn across account types and satisfaction levels.

a. In the table shown, an unexpected finding is recorded. Determine this finding that deserves further investigation. Suggest possible reasons for this finding.
b. Determine the highest churn rate and its corresponding category factors.

Question # 2

(3 Marks)

The figure below shows a histogram of review scores in the range 0-5 (discrete values) on Moon Time hotel services in Hurghada.

Histogram of Review Scores for Moon Time Hotel (Hurghada)

frequency

Review Score

a. Determine the mean and standard deviation of the above distribution.
b. Determine the Skewness. Comment on the calculated values.
c. Moon Time hotels conducted another survey to check its services in Sharm El Shiekh and Taba, th output results were: In Sharm El Shiekh the mean reviewers score was 4.20 with a standard deviatio of 0.05. In Taba, the mean reviewers score was 4.45 with a standard deviation of 0.5. Comment the performance on both hotels given the mean and standard deviation.

Q1-a) unexpected finding is that people with high satisfaction rates don't come back. Maybe subscribtion costs are too high

b) low, from Savings/Low

Q2) $N = 10+132+53 = 195$

$$\mu = \frac{(3\times10)+(132\times4)+(53\times5)}{195}$$

$= 4.22$ $\quad \sigma = \sqrt{\frac{\Sigma(x-\bar{x})^2}{n}} = 0.52$

b) left skewed → most of the scores are either 4 or 5 only 10 are 3. the number "10" relative to 50 and 130 is small, so the data will be skewed towards the left.

c) Sharm: $\mu = 4.2$, $\sigma = 0.05$ → rating is consistently around 4.2 due to a very low σ

Taba: $\mu = 4.45$, $\sigma = 0.5$ → rating is 4.45 but its not consistent given the very high σ meaning people give fluctuated reviews.

(5 Marks)

...#3

...tic company launches a new skincare serum and wants to analyze the relationship between sales ...e and four key factors:

- **Advertising Budget ($)** – The amount spent on marketing campaigns.
- **Social Media Engagement (%)** – The percentage increase in likes, shares, and comments.
- **Store Availability (Number of Stores)** – The number of retail locations carrying the product.
- **Customer Reviews (Average Rating out of 5)** – The average rating given by customers on e-commerce platforms.

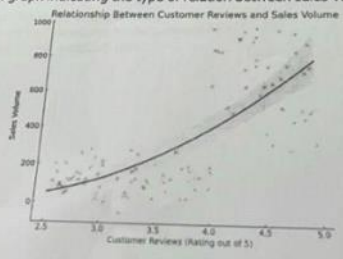
Factor	Sales Volume	Advertising Budget	Social Media Engagement	Store Availability	Customer Reviews
Sales Volume	1.00	0.85	0.75	0.80	0.65
Advertising Budget	0.85	1.00	0.70	0.60	0.50
Social Media Engagement	0.75	0.70	1.00	0.55	0.40
Store Availability	0.80	0.60	0.55	1.00	0.45
Customer Reviews	0.65	0.50	0.40	0.45	1.00

a. Store Availability has a correlation of 0.80 with Sales Volume, while Customer Reviews have a correlation of 0.65. What does this suggest about the relative importance of these factors?

b. The data shows a strong positive correlation (0.75) between Social Media Engagement and Sales Volume. Can we conclude that increasing the Engagement directly causes higher sales? Why or why not?

c. Suppose a scatter plot of Customer Reviews vs. Sales Volume shows that most data points cluster between 3.5 and 4.5 stars, but a few extreme cases (e.g., a product with a 1-star average and low sales) exist. How could these outliers affect the correlation coefficient? What statistical techniques can help mitigate this impact?

d. The following scatter plot further investigates the relation between sales and customer reviews, describe the shown graph indicating the type of relation between sales volume and customer reviews.



Q3-a) This shows a strong relation between Store availability and Sales volume, Same with customer Reviews but not as strong. this means that SA and SV are more strongly associated

b) No, because corelation is not causation meaning that, yes they are related, but this doesn't mean that they both cause a change in each other.

c) They can make the coefficient of correlation seem stronger or weaker than it is. removing errors or using methods like standardization like robust, yeoJohsnon, or z-score.

d) heteroscedastic graph meaning whenever the rating increases clusters become more scattered.

Choose the correct answer

1. Which transformation is capable of handling both positive and negative values?
   A. Min-Max Scaling
   B. Box-Cox
   C. Z-score Standardization
   D. Yeo-Johnson

2. Which scaling technique is sensitive to outliers? (2 Marks)
   A. RobustScaler
   B. Min-Max Scaler
   C. StandardScaler
   D. Normalizer

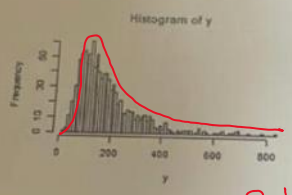
3. What does a heatmap represent in multivariate EDA?
   A. Category counts
   B. Frequency distribution
   C. Correlation between variables
   D. Outliers in data

4. Which scaler is most robust to outliers?
   A. MinMaxScaler
   B. StandardScaler
   C. Normalizer
   D. RobustScaler

5. Which plot is useful for showing correlation between many variables?
   A. Pair Plot
   B. Pie Chart
   C. Histogram
   D. Line Plot

6. What does a correlation coefficient of -0.95 imply?
   A. Strong positive correlation
   B. No correlation
   C. Strong negative correlation
   D. Moderate negative correlation

II. For the following figure (2 Marks)



a. Describe the distribution of the data in terms of skewness type → Righ Skewness
b. Identify two types of transformation techniques that fit it → log, robust, box-cox
c. Identify the problem in the provided data

skewness, because data is deviated